

**11** Changes in temperature can affect the spontaneity of a reaction.

Which statements are correct?

- 1 If a reaction is endothermic and has a negative entropy change, the reaction is spontaneous at all temperatures.
- 2 If a reaction is exothermic and has a positive entropy change, the reaction is spontaneous at all temperatures.
- 3 If a reaction is exothermic and has a negative entropy change, the reaction is **not** spontaneous at any temperature.

**A** 1 and 3

**B** 1 only

**C** 2 only

**D** 3 only